MTH166

Lecture-21

Partial Differential Equations (PDE)

Unit 4: **Partial Differential Equations**

(**Book:** Advanced Engineering Mathematics by R.K.Jain and S.R.K Iyengar,**Chapter-9**)

Topic:

Partial Differential Equations (PDE)

Learning Outcomes:

1. Formation of PDE by elimination of arbitrary constants

2. Formation of PDE by elimination of arbitrary functions

Partial Derivatives:

Earlier: If $\boldsymbol{u = f(x)}$, it means there is dependent variable *u* and one independent variable *x*.

So, we differentiate *u* with respect to *x* and denote it as: $\boldsymbol{\frac{du}{dx}}$

Now: If $\boldsymbol{u = f(x, y)}$, it means there is dependent variable *u* and two independent variable *x* and *y*. So, we can differentiate *u* with respect to *x* or y, denoted as: $\boldsymbol{\frac{\partial u}{\partial x}}$ or $\boldsymbol{\frac{\partial u}{\partial y}}$ respectively.

Standard Notations:

$$\left.\begin{aligned}\frac{\partial u}{\partial x} &= u_x = p \\ \frac{\partial u}{\partial y} &= u_y = q\end{aligned}\right\}$$ These are called first order partial derivatives.

$$\left.\begin{aligned}&\frac{\partial^2 u}{\partial x^2} = u_{xx} = r \\ &\frac{\partial^2 u}{\partial x \partial y}\ or\ \frac{\partial^2 u}{\partial y \partial x} = u_{xy}\ or\ \ u_{yx} = s \\ &\frac{\partial^2 u}{\partial y^2} = u_{yy} = t\end{aligned}\right\}$$ These are called second order partial derivatives.

Partial Differential Equation:

An equation of the form: $f(x, y, u, p, q) = 0$ is called first order partial differential equation.

An equation of the form: $g(x, y, u, p, q, r, s, t) = 0$ is called second order partial differential equation.

Methods of Formation of PDE:

1. By elimination of arbitrary constants

Problem 1. **Form the PDE for:** $\boldsymbol{u = ax + by,}$ $\boldsymbol{a}$ **and** $\boldsymbol{b}$ **are constants**

Solution. Given $u = ax + by$ (1) $[u = f(x, y)]$

Differentiating (1) partially w.r.t x

$$\frac{\partial u}{\partial x} = a(1) + b(0) = a \qquad \Rightarrow p = a$$

Differentiating (1) partially w.r.t y

$$\frac{\partial u}{\partial y} = a(0) + b(1) = b \qquad \Rightarrow q = b$$

Putting values of a and b in equation (1):

$u = px + qy$ **or** $u = \left(\frac{\partial u}{\partial x}\right)x + \left(\frac{\partial u}{\partial y}\right)y$ which is the required PDE.

<u>Problem 2.</u> Form the PDE for: $\boldsymbol{u = ax + by + a^4 + b^4}$, ***a*** **and** ***b*** **are constants**

<u>Solution.</u> Given $u = ax + by + a^4 + b^4$ (1) $[u = f(x, y)]$

Differentiating (1) partially w.r.t *x*

$$\frac{\partial u}{\partial x} = a(1) + b(0) = a \qquad \Rightarrow p = a$$

Differentiating (1) partially w.r.t *y*

$$\frac{\partial u}{\partial y} = a(0) + b(1) = b \qquad \Rightarrow q = b$$

Putting values of *a* and *b* in equation (1):

$$u = px + qy + p^4 + q^4 \qquad \textbf{or} \qquad u = \left(\frac{\partial u}{\partial x}\right)x + \left(\frac{\partial u}{\partial y}\right)y + \left(\frac{\partial u}{\partial x}\right)^4 + \left(\frac{\partial u}{\partial y}\right)^4$$

which is required PDE.

<u>Problem 3.</u> Form the PDE for: $\boldsymbol{u = (x-\alpha)^2 + (y-\beta)^2}$, $\boldsymbol{\alpha}$ **and** $\boldsymbol{\beta}$ **are constants**

<u>Solution.</u> Given $u = (x-\alpha)^2 + (y-\beta)^2$ (1) $\qquad$ $[u = f(x,y)]$

Differentiating (1) partially w.r.t x

$$\frac{\partial u}{\partial x} = 2(x-\alpha) \qquad \Rightarrow p = 2(x-\alpha) \qquad \Rightarrow \frac{p}{2} = (x-\alpha)$$

Differentiating (1) partially w.r.t y

$$\frac{\partial u}{\partial y} = 2(y-\beta) \qquad \Rightarrow q = 2(y-\beta) \qquad \Rightarrow \frac{q}{2} = (y-\beta)$$

Putting values of $(x-\alpha)$ and $(y-\beta)$ in equation (1):

$$u = \left(\frac{p}{2}\right)^2 + \left(\frac{q}{2}\right)^2 \qquad \textbf{or} \qquad 4u = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2$$

which is required PDE.

Polling Quiz

The PDE for function: $\boldsymbol{u = ax + by + a^2b^2}$, $\boldsymbol{a}$ **and** $\boldsymbol{b}$ **are constants, is:**

(A) $u = aq + bp + q^2p^2$

(B) $u = ap + bq + a^2b^2$

(C) $u = px + qy + p^2q^2$

Methods of Formation of PDE:

2. **By elimination of arbitrary functions**

Problem 1. **Form the PDE for:** $\boldsymbol{u = f(x^2 + y^2)}$

Solution. Given $u = f(x^2 + y^2)$ (1) $[u = f(x, y)]$

Differentiating (1) partially w.r.t x

$$\frac{\partial u}{\partial x} = f'(x^2 + y^2)(2x) \Rightarrow p = f'(x^2 + y^2)(2x) \Rightarrow \frac{p}{2x} = f'(x^2 + y^2) \qquad (2)$$

Differentiating (1) partially w.r.t y

$$\frac{\partial u}{\partial y} = f'(x^2 + y^2)(2y) \Rightarrow q = f'(x^2 + y^2)(2y) \Rightarrow \frac{q}{2y} = f'(x^2 + y^2) \qquad (3)$$

Comparing (2) and (3), we get: $\frac{p}{2x} = \frac{q}{2y}$

$\Rightarrow py = qx$ **or** $\left(\frac{\partial u}{\partial x}\right) y = \left(\frac{\partial u}{\partial y}\right) x$ which is the required PDE.

<u>Problem 2.</u> Form the PDE for: $\boldsymbol{u = f\left(\frac{x}{y}\right)}$

<u>Solution.</u> Given $u = f\left(\frac{x}{y}\right)$ (1) $[u = f(x, y)]$

Differentiating (1) partially w.r.t x

$$\frac{\partial u}{\partial x} = f'\left(\frac{x}{y}\right)\left(\frac{1}{y}\right) \quad \Rightarrow p = f'\left(\frac{x}{y}\right)\left(\frac{1}{y}\right) \quad \Rightarrow py = f'\left(\frac{x}{y}\right) \qquad (2)$$

Differentiating (1) partially w.r.t y

$$\frac{\partial u}{\partial y} = f'\left(\frac{x}{y}\right)\left(\frac{-x}{y^2}\right) \quad \Rightarrow q = f'\left(\frac{x}{y}\right)\left(\frac{-x}{y^2}\right) \quad \Rightarrow -q\,\frac{y^2}{x} = f'\left(\frac{x}{y}\right) \qquad (3)$$

Comparing (2) and (3), we get: $\frac{p}{2x} = \frac{q}{2y}$

$$\Rightarrow py = -q\,\frac{y^2}{x} \quad \Rightarrow px + qy = 0 \ \textbf{ or } \left(\frac{\partial u}{\partial x}\right)x + \left(\frac{\partial u}{\partial y}\right)y = 0$$

which is the required PDE.

Problem 3. **Form the PDE for:** $\boldsymbol{u = f(ax + by)}$

Solution. Given $u = f(ax + by)$ (1) $[u = f(x, y)]$

Differentiating (1) partially w.r.t x

$$\frac{\partial u}{\partial x} = f'(ax + by)(a) \qquad \Rightarrow \frac{p}{a} = f'(ax + by) \qquad (2)$$

Differentiating (1) partially w.r.t y

$$\frac{\partial u}{\partial y} = f'(ax + by)(b) \qquad \Rightarrow \frac{q}{b} = f'(ax + by) \qquad (3)$$

Comparing (2) and (3), we get: $\frac{p}{a} = \frac{q}{b}$

$$\Rightarrow pb = qa \quad \textbf{or} \quad \left(\frac{\partial u}{\partial x}\right) b = \left(\frac{\partial u}{\partial y}\right) a$$

which is the required PDE.

Polling Quiz

The PDE for function: $\boldsymbol{u = f\left(\frac{ax}{by}\right)}$, $\boldsymbol{a}$ **and** $\boldsymbol{b}$ **are constants, is:**

(A) $px = qy$

(B) $px + qy = 0$

(C) $py = qx$